FOUNDATION FOR FOOD & AGRICULTURE RESEARCH

FFAR Final Progress Report

As part of closing out a FFAR grant, grantees must complete a Final Progress Report. Please use the template below to complete the programmatic report. The report must be submitted to FFAR within 90 days after the expiration or termination of a FFAR funded research grant. All questions about this form should be directed to grants@foundationfar.org.

The Final Progress Report communicates the cumulative results and major accomplishments of the funded grant research, including accomplishment for each aim and whether all research aims were fully completed. FFAR uses the content of the Final Progress Report to inform Congress, the Advisory Board, and other stakeholders on the successes, impacts and value of funding unique partnerships to support innovative science addressing today's food and agriculture challenges. If a question is not applicable to the project, please type "Nothing to report."

Grant Information	
Grant ID	CA-PPP--0000000005
Award Program	Plant Protein Enhancement Program
Project Title	Using novel genes from wild germplasm to boost protein content in cultivated chickpea
Grant Start Date	08/01/2020
Grant End Date	07/31/2022
Total Grant Budget	$1,000,000
Total FFAR Budget	$
Matching Funder(s)	N/A
Final Report Date	10/31/2022
Project Director/Principal Investigator Information	
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Grantee Organization Information	
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General Information

1.1. Please list the geographic location(s) – city, state, congressional district - where the work was conducted. If the work was conducted outside of the US, please list the city and country.

The work was conducted in:

- Davis, CA, in California's 3rd Congressional district
- Sacramento, CA in California's 8th Congressional district

1.2. How many new jobs were created because of this grant funding.

• Direct: 13 jobs (including contractors) paid in whole or in part by FFAR grant funding.

• Indirect: 3 jobs (including contractors) paid by investment funding incidental to the FFAR grant funding.

1.3. How many jobs were maintained because of this grant funding?

• All jobs were created as a result of this grant.

1.4. Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant? If yes, please explain.

• No, we remain a for-profit Delaware c-corporation.

1.5. Has your organization undergone a recent name change? If so, please provide the new name of your organization.

• No

2. Scientific Report

2.1. Public Abstract (up to 500 words): The abstract should be 'stand-alone' and is intended for a general audience. It should be a concise overview/summary of the importance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of the project help to address important needs in U.S. food and agriculture systems.

A rapidly growing human population requires that we produce more food on less land, in an efficient and environmentally sustainable manner. Legumes provide a sustainable alternative to animal protein due to their reduced carbon footprint and high protein content. A symbiotic association with nitrogen fixing bacteria allows legumes to acquire reduced nitrogen in their roots which fuels protein biosynthesis. Chickpea already serves as a principal protein source for millions of people worldwide and is growing in popularity with food manufacturers as a plant-based protein ingredient. However, domestication reduced its genetic diversity, which has impeded efforts to improve chickpea's nutritional and agronomic qualities and limits potential as an alternative protein source. We have reintroduced genetic diversity into chickpeas by crossing elite cultivated varieties with a diverse collection of wild relatives to create a unique population with 40x more genetic variation. Several populations derived from these Wild x Cultivated (WxC) crosses produced seeds with 50% more soluble protein than modern, elite chickpea varieties. This observation led us to found NuCicer Inc., a company with the stated goal to produce high quality chickpea protein that is cost competitive with current plant-based proteins derived from soy and pea. It also served as the catalyst for the work described here, leveraging our unique genetic resource to revolutionize plant-based protein production in the U.S. and abroad by dramatically increasing the protein content of cultivated chickpeas. Over the two-year span of this FFAR funded project we have 1) Determined the protein content of >1,700 NuCicer chickpea varieties and characterized the protein profiles of a subset of these populations using quantitative proteomics; 2) Quantitated the starch and lipid content of a representative subset of chickpea lines and described the

relationships between these macronutrients and protein content; 3) Vastly improved the genome assemblies of cultivated chickpea *C. arietinum* and its wild progenitors *C. reticulatum* and *C. echinospermum*; 4) Identified nucleotide variation in gene-rich regions that differentiate the wild and cultivated genomes, from which we produced a high-throughput genotyping array; 5) Genotyped >3,500 unique NuCicer chickpea lineages, 1,700 of which have phenotypic data that were used to identify quantitative trait loci associated with protein content and key agronomic traits such as yield, growth habit, maturity date, and pod shattering; 6) Identified first generation F2-derived, high-protein chickpea lines with agronomic traits that are amenable to production at scale. Seed from these lines have been advanced to the 7^{th} filial generation ("F7") and increased to allow for protein isolation and characterization by our industrial partners; 7) Leveraged the genetic and phenotypic data generated from this project to inform crosses intended to further improve the nutritional and agronomic qualities of our chickpeas. In summary, with the help of FFAR, we have developed first generation, high yielding, high-protein lines that provide the basis to efficiently produce a high quality, plant-based protein. Furthermore, we have developed genomic resources that will allow us, and other chickpea breeders, to further improve the crop, making it a viable protein option in the US Food production system.

2.2 Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making (up to 1,000-word limit).

NuCicer's high protein chickpea seed, flour and protein-enriched ingredients are unique and create new opportunities at multiple points in the value chain. At the farm-level, NuCicer is working with farming logistics companies and commercial growers to both test and increase seed for NuCicer genotypes. Success means that NuCicer chickpeas will be grown at commercial scale, primarily in the North Central and Pacific Northwest farming regions. The outcomes of commercial production will connect NuCicer's chickpea protein products to NuCicer's customers who are consumer packaged goods (CPG) companies and companies positioned intermediate in the value chain. Ultimately, NuCicer protein ingredients will impact consumers in terms of improved nutrition and health, reduced cost, and new manufactured food products that involve plant-based protein.

2.3 Research Outcomes/Impact (up to 1,500-word limit). The primary goal here is to answer questions such as "How did this project lead to improvements in U.S. agriculture and food systems?" or "How can the findings of this study guide or make improvement in future investigations and research?" Outcomes should be explained and classified in one of the following ways:

a) potential outcomes, i.e., findings, results, or recommendations that could impact U.S. agriculture and food systems, if used;
b) b. intermediate outcomes, i.e., how the findings, results, or recommendations have been used by others to influence practices, legislation, research/product design, training and so forth; and
c) c. end outcomes, i.e., how findings, results, or recommendations have contributed to documented reductions in work-related morbidity, mortality, and/or exposure.

Potential impacts on US agriculture and food systems, beyond those described above (2.2), include (1) more secure on-farm income through the use of contracted production, (2) reduced environmental impact based on reductions to nitrogen fertilizer inputs, (3) reduced water inputs based on chickpea's inherent water-deficit resiliency and the potential of NuCicer genetics to further improve this trait, (4) lower cost chickpea protein products, which will increase access to protein ingredients with preferred functionalities.

2.4 What were the goals/specific aims of the project? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets. (up to 1,500-word limit)

1. Finish developing F5 RIL population and develop down-selected F6 RIL population (Completed: September 2021)

1.1 Advance 620 F5 lines to F6 in 50 plant subplots. (Completed: November 2020)
1.2 Advance 1,800 F4 lineages to F5 (Completed: September 2021)

2. Determine the range of seed protein content and seed composition (Completed: July 2022)

2.1 Determine total protein content (fluorescent assay) and total starch content (enzymatic assay) across F5 and F6 material. (Completed: July 2022)

2.2 Submit 100 selected lines from the F5/F6 material for TMT MS, DUMAS, and seed characterization testing. (Completed: December 2021)

3. Genotype germplasm (Completed: June 2022)

3.1 Develop Illumina Infinium iSelect Array for chickpea genotyping. (Completed: December 2020).

3.2 Genotype 2,300 lines using the developed iSelect array (Completed: June 2022)

4. Determine genetic control of seed protein content and quality (Complete: July 2022)

4.1 Utilize seed protein values and genotype data to identify loci controlling seed protein content via multi-QTL analysis (Complete: July 2022).

4.2 Test for interactions between seed protein QTL and other agronomic traits (Complete: July 2022).

4.3 Develop genomic selection and speed breeding strategies (Complete: July 2022)

5. Develop next generation germplasm containing high seed protein content (Complete 2022).

5.1 Initiate genetic crosses to increase frequency of desirable alleles for protein and agronomic traits (Complete: May 2022).

5.2 Grow and share promising lines with partners for internal testing and analysis (Complete: July 2022).

5.3 Incorporate partner data and feedback into a variety development process (Complete: July 2022).

6. Submit selected information to a public-facing database. (Complete: June 2022).

6.1 Release selected genomic data to the National Center for Genome Resources for public availability. Improved genome assembly, gene prediction and annotation. (Complete: July 2022).

6.2 Upload selected F4 phenotype data to Chickpea Germinate. (Complete: August 2020).

2.5 Have any of the major goals/specific aims or milestones for the project changed since the award? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (up to 500-word limit)

No changes have been made to the goals for this project.

2.6 What was accomplished under the goals/specific aims or milestones for the project? Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (both positive and negative); and 4) key outcomes or other achievements. Include discussion of stated goals not met. The emphasis in reporting in this section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical as the PI would like. (up to 5,000-word limit)

Final Project Update:
Major activities, accomplishments, and results from the second half of year two are listed below followed by a summary of the accomplishments, key findings, and outcomes for each of the six objectives of the project.

Six-Month Update Summary
In the final six months of the project, we have completed our analysis of the protein of the 2021 field trial and initiated work to measure lipid and starch content. We regressed our quantitative proteomic data with both macronutrient composition and plant pedigrees to infer potential relationships between these datasets. 3,500 lines have been genotyped and QTL's related to protein content and key agronomic traits such as yield and growth habit identified. Genetic and phenotypic data were used to identify parents for crosses to combine beneficial traits, and new F1 individuals were generated. We finished harvesting our third field season and selections have been made for large scale grow-outs in winter nurseries.

Overall Accomplishments
Since the start of this FFAR funded project, we have advanced a diverse collection of 4,000 F3 and F4 populations into a well-characterized collection of F6-F8 populations with comprehensive genotypic, phenotypic, and macronutrient composition data. We have down selected first generation, high protein cultivars that we are bulking for industrial scale protein extraction and functional testing. Furthermore, we have utilized the data from this project to select parents for crosses to combine and further improve chickpea agronomic and macronutrient traits. The QTL's we identified during this project will accelerate these improvements. Additionally, during this process have developed several key genetic and genomic resources including vastly improved genome assemblies that will benefit not only NuCicer, but chickpea researchers, breeders and growers worldwide. In summary, this project has moved chickpea forward to make it a viable and desirable alternative to pea and soy protein as a key source of plant-based protein.

In the following paragraphs you will find additional details of our progress and accomplishments to date.

1. Finish developing F5 RIL population and develop down-selected F6 RIL population

A principal objective of this FFAR funded project was to advance our RIL populations to reduce heterozygosity, generate material for laboratory testing, and to collect agronomic data. Over the course of this project, we have advanced lines as many as four generations beginning with F4 seed in 2020 and collecting F8 seed from select lineages in our recent 2022 field experiments. We completed our stated goals of advancement to F5 in 2021 which brought our total F5 germplasm library to just under 4000 populations. More recently, we completed our 2022 summer grow-outs to further advance down selected lines to F6 - F8, to further reduce heterozygosity, test the stability of the high protein trait, and to increase seed quantity for large scale testing. Eleven lines of interest, selected for their high protein content and desirable agronomic traits, were grown in two independent locations yielding several hundred pounds of seed each. This material is currently being shipped to NuCicer partners for functional testing and additional scale up in winter nurseries.

2. Determine the range of seed protein content and seed composition

Seed coat and Protein Content.
Seed coat thickness is an important agronomic trait which impacts both seed processability and the amount of protein a plant produces per acre. In a laboratory setting, seed coat

removal is a rate limiting step in generating chickpea flour for compositional analysis. We therefore measured the relative contribution of seed coat to total seed biomass and used this value to extrapolate protein content measured on flour made from non-dehulled seed. In the fourth quarter of this project, we completed both our seed coat and protein analysis of the ~1600 F5 seed harvested in 2021. The relative contribution of seed coat to total biomass varied widely, ranging from 5-28%. The distribution in soluble protein content in flour made from non-dehulled seed was 10-26%. When corrected for the contribution of the seed coat, the soluble protein content of flour made from dehulled seed would be 11-31%, a range consistent with the results of our 2020 field trial. Prior experiments comparing the soluble protein content (i.e. extracted with 0.1 M NaOH) with the total protein content of flour as measured using DUMAS found that the soluble protein values represent approximately ~80% of the total protein in the chickpea flour. When the soluble protein values were converted to an estimated total protein, we calculated a range of 12-38% protein across the F5 seed produced in our 2021 trial. These values surpass protein values reported for field peas and approach or equal the protein values reported for soy, the two most common sources of plant-based protein. F4 populations from the 2021 field season found to contain high protein content and/or thin seed coats were selected for advancement in the 2022 grow-out. Notably, we identified several populations with one or both traits that also demonstrated upright growth, good yield values, and little to no shattering. These populations are of particular interest as they may already be commercially viable and are being further scaled up for additional analyses.

Starch. The relative abundance of starch will impact how our chickpea flour functions in specific food applications. Likewise, the level of resistant vs soluble starch has important health ramifications as resistant starch is reported to have added health benefits, particularly for the gut microbiome and colon health. At the time of our last update, we had submitted 22 samples representing a range of protein content to the UCD Analytical Lab for starch and total carbohydrate analysis. These analyses found that starch content varied widely from 31-44%, across all samples and was negatively correlated with protein content. We subsequently developed the capacity to perform our own starch analysis in-house and are working to collect sufficient data to map loci regulating starch content. We recently measured both the soluble and resistant starch from 60 individuals from 20 populations (3 siblings per population). Resistant starch quantities were significantly lower than soluble starch typically constituting between 1-3% of the sample biomass. Resistant, soluble, and total starch values were consistent between siblings and total starch was again negatively correlated with protein content. While the current dataset is not large enough to map loci regulating starch content, additional analyses are ongoing. We have recently made changes to the protocol to increase throughput and sample size to facilitate mapping efforts. Concurrently, we are working with an industrial food manufacturing partner to determine how different protein profiles and protein:starch ratios affect chickpea flour and chickpea protein extract functionality. The current starch and protein datasets were used to identify samples with diverse compositions for testing.

Lipid. Although chickpea contains significantly less oil than soybean and peanut, it has a higher fat content than pea and contains sufficient oil to limit how it can be processed or used. High fat flour can foul the membranes and pipelines used in protein extraction and can negatively impact shelf life and protein functionality in food applications. Just prior to our last research update, we submitted forty flour samples to the UC Davis Analytical Lab to survey the crude fat and total protein content in our WxC chickpea collection. These samples were previously analyzed by quantitative mass spectrometry and represent diverse pedigrees and soluble protein contents. As observed previously, the total protein values were highly correlated with our measurements of soluble protein and ranged from 18.6-36.6% (R^2=0.96). Notably, the crude fat content varied more than two-fold and ranged from 2.8-7.1% of total flour mass. As with the starch content, we observed a negative correlation between total protein and crude fat content. High protein chickpeas contain less fat than low protein chickpeas (R^2=0.62). We also characterized the fatty acid profile of 20 chickpea populations in collaboration with an Ag-Biotech partner. Fatty Acid (FA) Methyl Ester (FAME) analysis indicated that in all 20 samples, linoleic acid was the predominant FA (51-66% of total FA) followed by oleic (16-34%), palmitic (10-14%), and linolenic acid (2-4%). These data are consistent with published chickpea FA profiles. Total FA levels (w/w) varied from 2.7-5.1% and as with total fat and starch, were negatively correlated with protein content. Future work will use and advance these results with the goal of improving chickpea processability and functionality by controlling lipid content.

Proteomics. Since our previous update, quantitative proteomic data for a second plate of 96 samples from the UC Davis proteome center data have been received and analyzed. This brings the total to 192 protein samples across 60 populations run at 3 biological replications. These populations span the breadth of protein content and all cultivated and wild parents from our collection are represented at least once. Several identical samples were included on both plates which were used to normalize the data across the two experiments. >3,000 proteins were quantitated across the two datasets and while the quantity of most proteins were similar, specific proteins varied widely. Consistent with prior published descriptions of seed proteomes, seed storage proteins (SSP's) are among the most highly abundant proteins across all samples, and several were predictive of total protein content. Notably, proteins involved in starch and lipid biosynthesis and storage (e.g. starch synthase, oleosin) are also highly abundant and their relative expression is negatively correlated with several of the seed storage proteins and protein content more broadly. These observations agree with the results of our analytical experiments in which protein content was negatively correlated with both starch and oil.

Pedigree and protein content are two highly significant variables explaining much of the variation in protein profiles observed in our proteome data. Measured soluble protein content accounted for 22% of the proteome variation in an unsupervised PCA with high soluble protein (26-30%), medium soluble protein (19-25%) and low soluble protein (14-18%) samples forming clusters. Pedigree accounted for approximately 12% of the observed variation in a supervised PCA with data points from each of the four cultivated parents forming organized clusters. The contribution of the wild parents, collectively, to the protein profiles is less evident, although a few wild parent families do form distinct clusters. This is likely due to the larger number of highly diverse wild parents represented in the collection, which reduces statistical power. Nevertheless, the data support the conclusion that genes contributed by the wild parent affect the protein profile. We predict that the differences we observe in protein profiles will reflect differences in the functional properties of protein isolates, which is currently under investigation with commercial partners. The similarity of the protein profiles among individuals with similar pedigrees is therefore highly significant in that it implies a genetic component to the protein profiles and suggests that differences in functionality may also be heritable. Experiments characterizing the functionality of the extracts produced from these populations were recently initiated and we will be testing this hypothesis in the future.

In summary, we have used this project to thoroughly characterize the macronutrient composition of our unique chickpea collection. We have identified hundreds of populations with protein contents that are significantly higher than common cultivated chickpeas and have observed how high protein content impacts other nutrients such as starch and fat. We have used and continue to use these data to identify genetic markers that associate with each of these traits, which will in turn inform breeding strategies.

3. Genotype germplasm

In the past six months of this project, we have acquired Illumina iSelect genotyping data for the 1,100 F5 lineages advanced during our 2021 field season for which we have phenotypic data including yield, protein quantities. We have also generated Illumina data for an additional 1,800 F5 populations. Along with the initial 614 populations we initially genotyped, this brings the total number of genotyped populations to over 3,500.

4. Determine genetic control of seed protein content and quality.

Multi-QTL analyses and genome-wide association studies (GWAS) were conducted and revealed significant genetic loci for a variety of traits, including two significant loci of wild-origin that control protein quantity. Three additional loci are evident in the data but given constraints of population size their effects to not pass significance thresholds. Addition of the on-going data for 1,100 F5 lineages will contribute additional genetic and phenotypic data that should add additional power to the analysis.

In addition to protein quantity, we also identified significant loci controlling yield traits, plant architecture and seed coat characteristics. Although we observe interactions among some of these non-protein loci, we observe no interaction with protein content.

5. Develop next generation germplasm containing high seed protein content.

Selected high-protein F2 derived, F7 seed were scaled up in the Spring/Summer of 2022 to generate material for commercial trials. In parallel, genetic crossing strategies are being employed to develop second generation, high-protein, elite chickpea varieties. Parents were selected for new crossing matrices based on models generated from the recently completed genotyping project and the phenotypic data collected from the 2021 field season. New F1 individuals were derived using these crossing matrices as part of this grant, and we are moving forward with further advancements post-grant. Our speed breeding facility has been designed and is currently under construction. Once operational, this facility will significantly reduce seed-to-seed generation times and our breeding program will rapidly advance these materials to the field for trials.

In summary, over the two years of this project we have successfully built the genetic and genomic infrastructure to make significant increases in chickpea protein content. Genotypic and phenotypic data have informed preliminary selections for parents that will serve as the basis for initial breeding efforts and a crossing matrix has been constructed and initiated resulting in current F1 individuals. Although COVID-19 related delays in genotyping and construction issues have slowed progress, the process is gaining momentum and will continue to accelerate as these hurdles are overcome.

6. Submit selected information to a public-facing database. (Complete)

Phenotyping data is now available at Germinate Chickpea, which is a chickpea web site describing recombinant inbred lines up to F4 generation. In addition, we developed a web browser and data mining tools for PacBio HiFi genome assemblies. The genome data is freely available through the National Center for Genome Resources. A workshop introducing the web database to the international chickpea community is scheduled for November 3, 2022.

2.2. Discuss efforts taken to ensure the approach is scientifically rigorous. Include approaches taken to ensure robust and unbiased results.

NuCicer's company ethos is to use rigorous experimental design and statistical analyses to interpret and inform our research and development (R&D) activities. We have trained PhD scientists leading each of our R&D efforts, and close coordination between our lab/field teams and our computational/statistical teams, which together inform management and business strategy decisions. We have taken multiple steps to ensure that our conclusions are robust and unbiased. 1. Our agronomic and protein data have been collected over multiple growing seasons and from different growing environments. 2. Biological replicates grown in a common garden were analyzed to assess variation. 3. Samples were tracked using barcoded identification numbers to ensure anonymity. 4. Key experiments have been replicated by third parties (by collaborators or fee-for-service providers) and were run in an anonymized manner. 5. Field experiments included elite, cultivated chickpea varieties which served as important checks to benchmark both agronomic and nutritional profiles. 6. Common "control" samples were included in all protein and starch assays to assess batch-to-batch variation and to ensure the integrity of the commercial kit components. It is in NuCicer's best interest to ensure that our observations are robust and reproducible across both time and space and we feel these measures accomplish that goal.

2.3. List key stakeholders that could be served by results of this project.

The results of this project will benefit stakeholders throughout the value chain from the grower to the end consumer. Chickpea researchers both in academia and in the private sector have more robust genome assemblies to facilitate molecular and genetic analyses. Growers will benefit from our high yielding chickpea varieties that produce more protein per acre and thus command a premium price at market. More efficient chickpea protein production will bring costs down for food manufacturers and provide a cost-effective alternative to soy and pea protein in plant-based protein applications. Chickpea's mild flavor and high consumer acceptance will stimulate innovation by chefs and food scientists meaning that consumers will have more products, and higher quality products to choose from as they move away from animal-based products towards those made with plant-based ingredients.

2.4. How have the results of this reporting period been disseminated to communities of interest? Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write "nothing to report." A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sources will be reported under Information Products. (up to 1,000-word limit)

Our primary interaction with external communities is the development and release of new and improved genomic resources. In addition to high quality genome assemblies, we have determined gene structure based on RNAseq data and we have developed evidence-based annotations of three *Cicer* genomes. The data is served from a public JBrowse web site at the National Center for Genome Resources (http://dev.lis.ncgr.org:50024/cicer/?config=cicer%2Fconfig.json), and includes a number of precomputed data sets that add value to users (such as cross-species synteny analyses and phylogenetic trees), as well as query tools based on BLAST (http://dev.lis.ncgr.org:50018/blast/) and keyword searches.

2.5. Describe challenges or delays encountered during project and actions that were taken to resolve them. Only describe significant challenges that may have impeded the research and emphasize their resolution. (up to 500-word limit)

The primary delays that we experienced were related to work that was outsourced to UC Davis.

Genotyping that was performed by the UC Davis Genome center was delayed due to the backlog of COVID-19 samples they were running for the City of Davis. We initiated conversations with alternative genotyping service providers (Illumina) but this was terminated when UCD was eventually able to start work on our project. Ultimately this situation caused an

8-week delay in genotyping NuCicer's samples. While our team was still able to complete initial genetic modeling with results by the end of this reporting period, this delayed decisions for generating next generation germplasm

COVID related staffing issues also impacted work at the UCD Analytical lab, prolonging turnaround time for starch and lipid analysis. Although the Analytical lab eventually returned our data to us, we elected to develop our own in-house starch and lipid assays to expedite subsequent analysis.

2.6. What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to provide training and professional development during this period, state "Nothing to Report." For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals. (up to 500-word limit).

Nothing to report

2.7. Please indicate the number of undergraduate and graduate students, post-doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail. (up to 500-word limit)

Over the course of the project we employed five undergraduate students working as interns. Two students were trained on conducting fluorescence protein assays, four students have learned general lab techniques and greenhouse maintenance practices, and two students gained experience with field work.

3. Information Products

3.1. Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video, audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during the project resulting directly from the FFAR award.

In coordination with FFAR in August 2021 and then through a second press release in December 2021, NuCicer reported successfully increasing protein content in chickpea by 75%. This has generated many discussions about the work NuCicer is doing under the FFAR grant across the industry - from breeders to farmers to ingredient processors and food and beverage manufacturers.

3.2. Please provide a list of citations for the information products produced during the project.

Nothing to report

3.3. Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the project resulting directly from the FFAR award? If yes, please provide citation.

Nothing to report

3.4. Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.

Public database for Cicer genomic resources. This web site integrated genome sequence, gene expression, gene annotation, cross-species synteny and phylogenetic data.
http://dev.lis.ncgr.org:50024/cicer/?config=cicer%2Fconfig.json

3.5. Have inventions, patent applications, and/or licenses resulted from the project (U.S. and international)? If yes, indicate the invention, patent application(s), and/or license(s).

	# Filed (Enter Numeric Value)	# Approved (Enter Numeric Value)	Patent Numbers (Separated by commas)
Patents (Provisional)	1	#	Application No. 63/411,236
Copyrights	#	#	
Trademarks	#	#	

Inventions	#	#	[redacted]

3.6. Are any of the information products produced during this grant that are confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if they must remain confidential indefinitely. (up to 500-word limit).

- All data, images, models, and all other materials produced during the reporting period of this grant are considered confidential NuCicer proprietary information.

- Per the grant agreement, at the conclusion of the grant period NuCicer will develop an approach to ensure selected results and data from the grant are available to the scientific community in an appropriate and timely manner. The public workshop on genome resources, scheduled for November 3 over Zoom, is an example of this outcome.

3.7. Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers, and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that other can access and re-use these data in the future?

The database that we have helped develop is an interactive tool that facilitates data mining from the Cicer genome. It has utility for both breeders and researchers and was designed with end-users needs in mind.

4. Data Management

4.1. Did the project generate any data? Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and imageries). Please list the data generated for the award.

1 Genome assembly for *C. arietinum* was significantly improved. Two *de novo* genomes for *C. reticulatum* and *C. echinospermum* were developed.

Proprietary data include:

-5,000 SNPs were identified across the genomes and used to construct an Illumina iSelect array.
-Illumina Genotyping data was collected for >3,500 unique populations
-Quantitative Proteomic analysis was run on 192 protein samples representing ~60 unique populations.
-Soluble Protein content values were determined for the flour generated from ~1,700 individual lines. Most values were obtained by measuring extracted protein with a commercial kit while DUMAS was also used to calculate total protein for ~600 populations. A subset of these lines were advanced during 2022 in three different growing locations and soluble protein content was determined for these samples.
- The percentage of seed biomass contained in the Seed Coat vs the cotyledons was determined for 1,100 F5 individuals.
-Seed shape, Seed size, seed color, Plant Yield, Plant Growth Habit, Plant biomass, Harvest Index, Harvest date/maturity, seed shattering data were collected for 620 individual populations during the 2020 field seasons. These data were collected for 1,100 populations during 2021 in addition to seeds per pod, pods per node, and seed coat biomass.
-Soluble and Resistant Starch values were determined for a subset of samples from the 2020 and 2021 field seasons, and the 10 most advanced lines grown in 2022
-Fatty Acid (FAME), Amino Acid, and starch data was also collected for 20 lines in a collaboration with an industrial partner.

4.2. If you list multiple data sets, are these data sets related? If so, please provide a short description of how they are related.

All data sets were collected from observations made or samples taken on/from individual populations from the same collection. The same populations were often observed over multiple generations, and from the same generation grown in different environments.

4.3. Please provide copies of relevant metadata records to support FFAR's mission of enhancing the discoverability of FFAR funded project data and information products. Include or attach copies of records and a simple file inventory, if necessary, in a compressed folder.

The current web address for Cicer genome resources is (http://dev.lis.ncgr.org:50024/cicer/?config=cicer%2Fconfig.json). Our partners at the National Center for Genome Resources are currently develop a home page for the project, which will enhance the ability of users to navigate the data.

FOUNDATION FOR FOOD & AGRICULTURE RESEARCH

Certification

The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate and complete. I am aware there is a significant penalty for submitting false or misleading information.

If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.

PI Full Name: Brendan Riely

PI Signature: ____________ Date: October 31, 2022

Authorized Signing Official (ASO) Name: Kathryn Cook

ASO Signature: Kathryn Cook Date: October 31, 2022